ARCHITECTURE BRIEF · APRIL 2026

Network Security in Microsoft Fabric.

Three pillars. Zero public exposure. One coherent architecture for enterprise-grade data platforms.

FREDGIS · GITHUB.COM/FREDGIS/FABRIC-FOUNDRY-KB/

The Shift

Fabric is **SaaS**. The public endpoint **is** the platform.

Network security becomes a **composition of controls** — each addressing a threat surface, none of them sufficient alone.

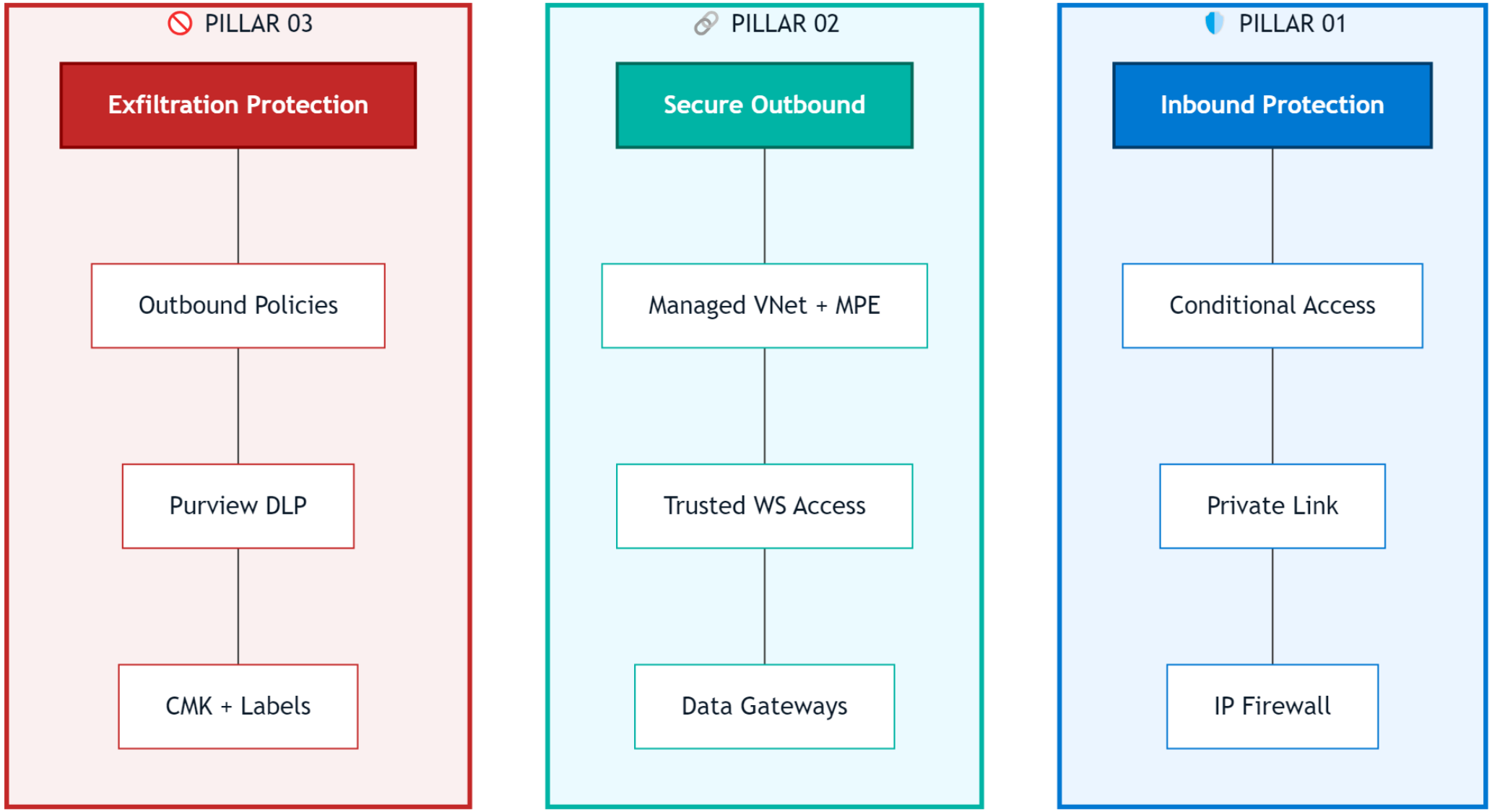
The goal: **layered containment** that assumes breach at every tier.

3 pillars of network defense

0 trust assumptions — Zero Trust by design

1 coherent DEP architecture

The Three Pillars



Each Pillar Answers a Different Question

PILLAR 01

Inbound Protection

Who can access Fabric?

Conditional Access | Private Link | IP Firewall

PILLAR 02

Secure Outbound

How does Fabric reach private sources?

Managed VNet | MPE | TWA | Gateways

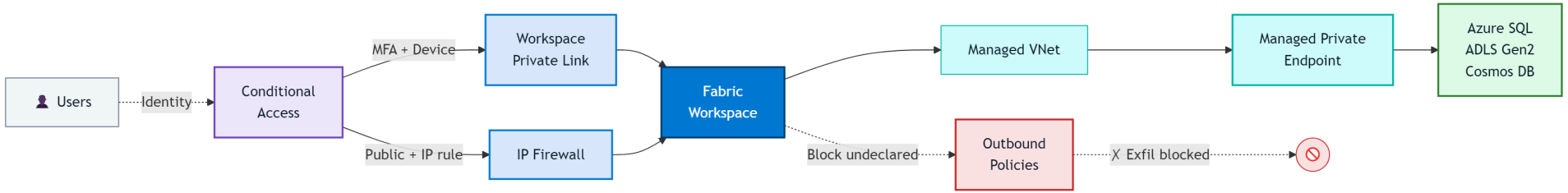
PILLAR 03

Exfiltration Protection

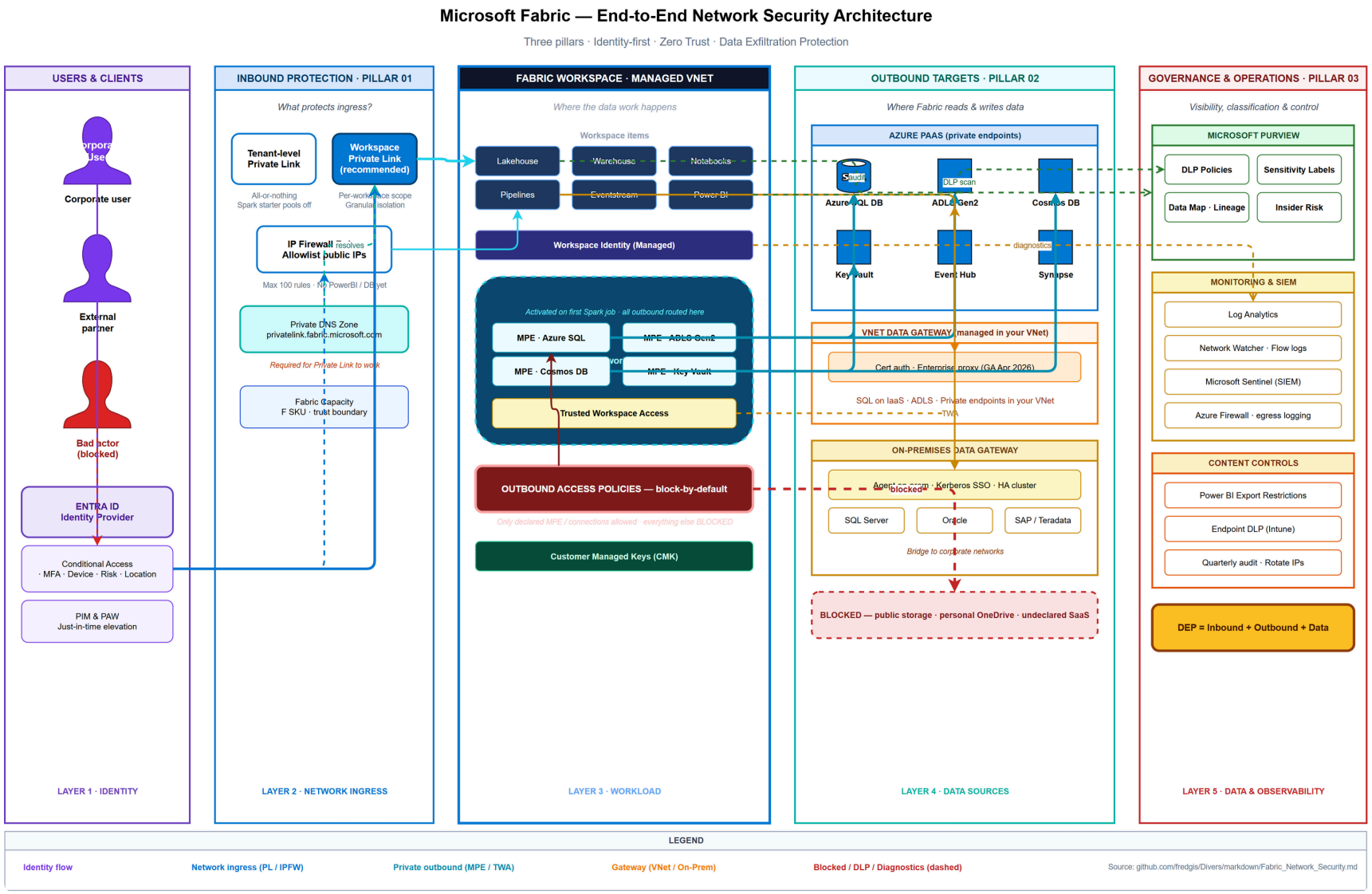
Where can data go?

Outbound Policies | Purview DLP

End-to-End Flow



Reference Architecture



01

Inbound Protection.

CONTROLLING WHO — AND FROM WHERE — CAN REACH THE FABRIC TENANT.

Conditional Access

POLICIES EVALUATE

- **User / Group** · per business unit
- **Location** · untrusted countries
- **Device** · Intune compliance
- **Risk level** · P2 licence
- **App** · per-client granularity

ENFORCE

MFA | Session lifetime | PIM

REQUIREMENTS

Component	Level
Entra ID	**P1**
MFA	**Required**
Device compliance	Intune
Risk-based CA	**P2**

> **Rule of thumb.** If CA isn't configured, nothing below matters.

Private Link — Tenant vs Workspace

TENANT-LEVEL

All-or-Nothing

Single service for the entire tenant.

Simpler deploy | Spark starter pools off | Some features unsupported

WORKSPACE-LEVEL · RECOMMENDED

Surgical Isolation

One service per workspace. Isolate only what needs it.

Per-workspace scope | Share a VNet | Tenant admin toggle

> Both require a **Private DNS Zone** (`privatelink.fabric.microsoft.com`). Forgetting it is the #1 reason PL "doesn't work" on first deployment.

IP Firewall Rules

Allow Fabric on the public internet — but only from **declared IP ranges**. No VNet needed.

Limits

- No Power BI items yet
- No Fabric databases yet
- REST API always reachable
- Max **100 rules** per WS

SUPPORTED ITEMS

Lakehouse | Warehouse | Notebook | Pipeline | Dataflow Gen2 | Eventstream | Mirrored DB

COMBINES WITH

Private Link + IP Firewall → private paths and allowed public IPs both permitted. Everything else denied.

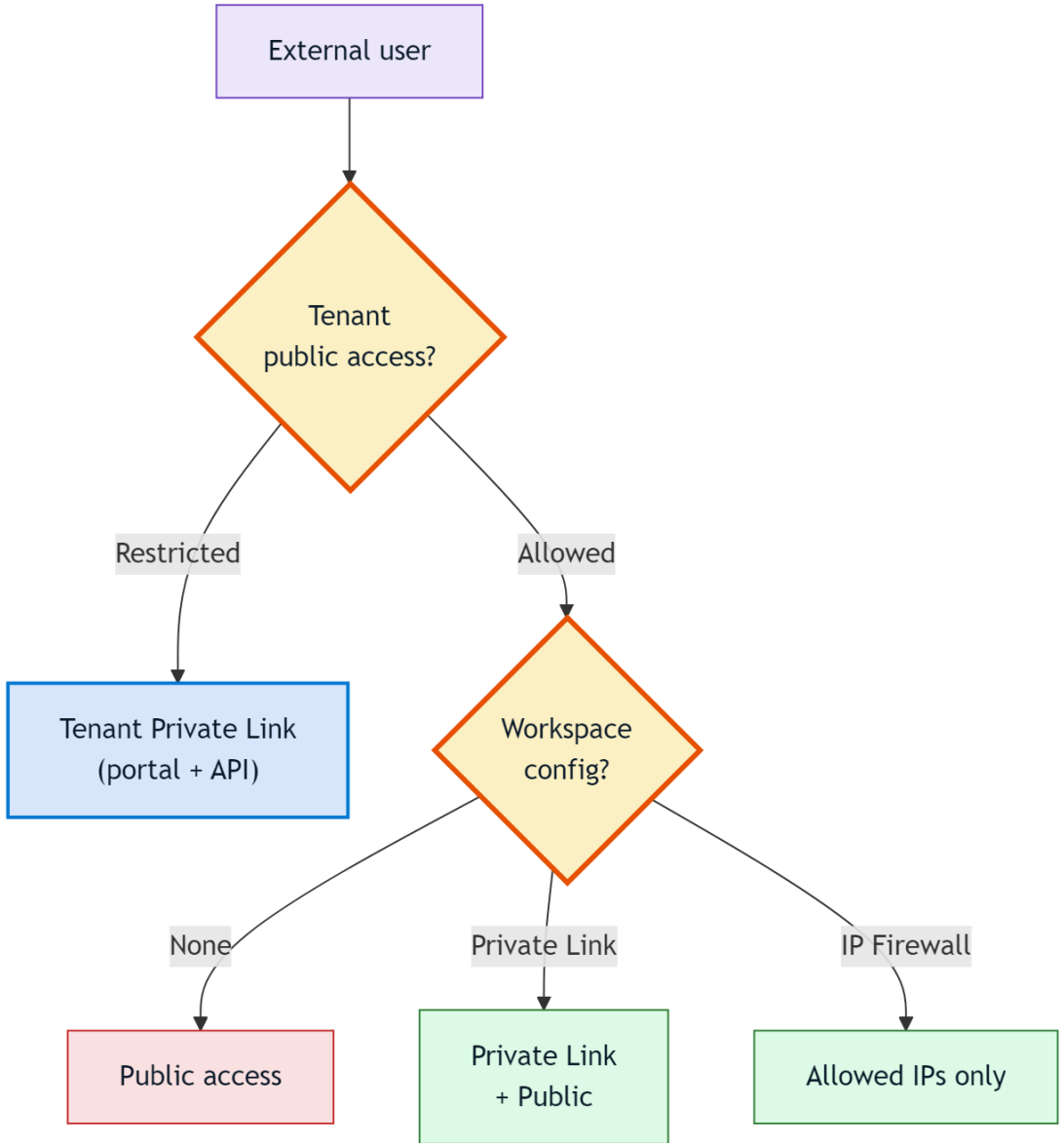
GA since early 2026

Tenant × Workspace Interaction

Two knobs, combined:

- **Tenant** `Allowed` or `Restricted`
- **Workspace** `None`, `PL`, or `IP Firewall`

Non-obvious: when tenant public access is restricted, workspace-level PL enables **API access only**. Portal needs tenant-level PL.

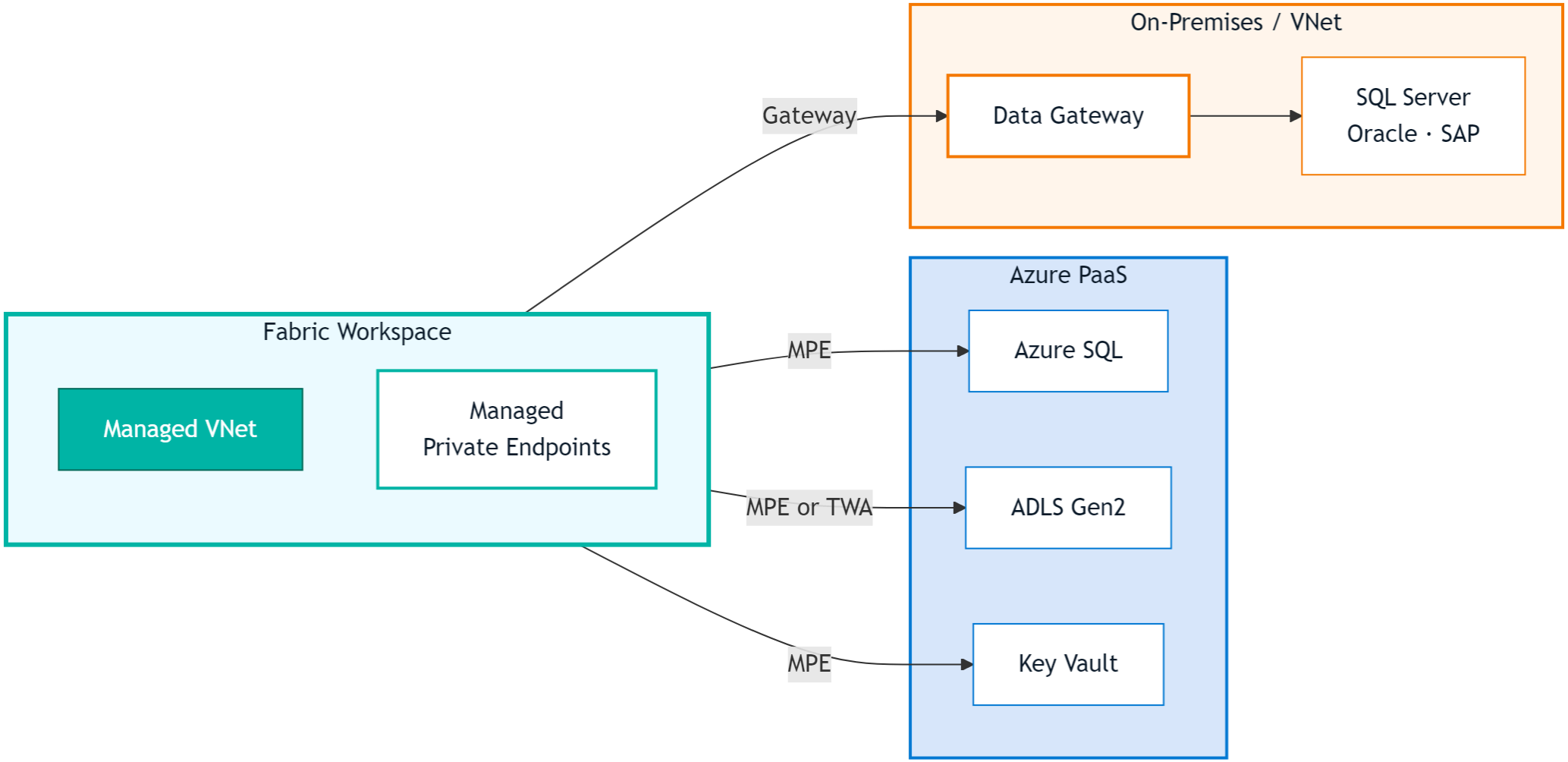


02

Secure Outbound.

REACHING PRIVATE DATA SOURCES — WITHOUT CROSSING THE PUBLIC INTERNET.

Outbound Architecture



Managed VNet & Private Endpoints

MANAGED VNET

- **Microsoft-managed** per workspace
- Activated on first Spark job
- All outbound routed through it

MANAGED PRIVATE ENDPOINTS

- Private connections to Azure PaaS
- Traffic on Microsoft backbone
- Target service can block public access

SUPPORTED TARGETS

Azure SQL DB | ADLS Gen2 | Cosmos DB | Key Vault | Synapse | Purview | Event Hub

> **MPE flow.** Create → target admin approves → private traffic flows. Approval is intentional — prevents bypass.

Trusted Workspace Access

TWA reaches **firewalled ADLS Gen2** without deploying a Private Endpoint.

Useful when MPE is overkill — single target, simple RBAC.

PICK TWA OVER MPE WHEN

- Target is **ADLS Gen2 only**
- No VNet infra desired
- Speed > isolation purity

PREREQUISITES

01 F SKU capacity
Trial / PPU don't qualify

02 Workspace Identity
Generated by admin

03 Storage RBAC
Storage Blob Data Contributor

04 Resource Instance Rule
On the storage firewall

Data Gateways

ON-PREMISES DATA GATEWAY

Bridge to corporate networks

Agent on a server inside your network. Encrypts outbound to Fabric.

✓ SQL Server, Oracle, SAP, Teradata
✓ Clustering for HA
✓ Kerberos SSO supported

VNET DATA GATEWAY

Managed gateway in your VNet

No VM to maintain. Fully managed, deployed in your VNet.

✓ SQL on IaaS, private endpoints
✓ **Certificate auth · GA**
✓ **Enterprise proxy · GA**

Outbound Connector Matrix

Target	MPE	TWA	On-Prem GW	VNet GW
ADLS Gen2	✓	✓	—	✓
Azure SQL	✓	—	✓	✓
Cosmos DB	✓	—	—	✓
Key Vault	✓	—	—	—
SQL Server IaaS / on-prem	—	—	✓	✓
Synapse	✓	—	—	✓
Purview	✓	—	—	—
Event Hub / Service Bus	✓	—	—	✓
SAP / Oracle on-prem	—	—	✓	—

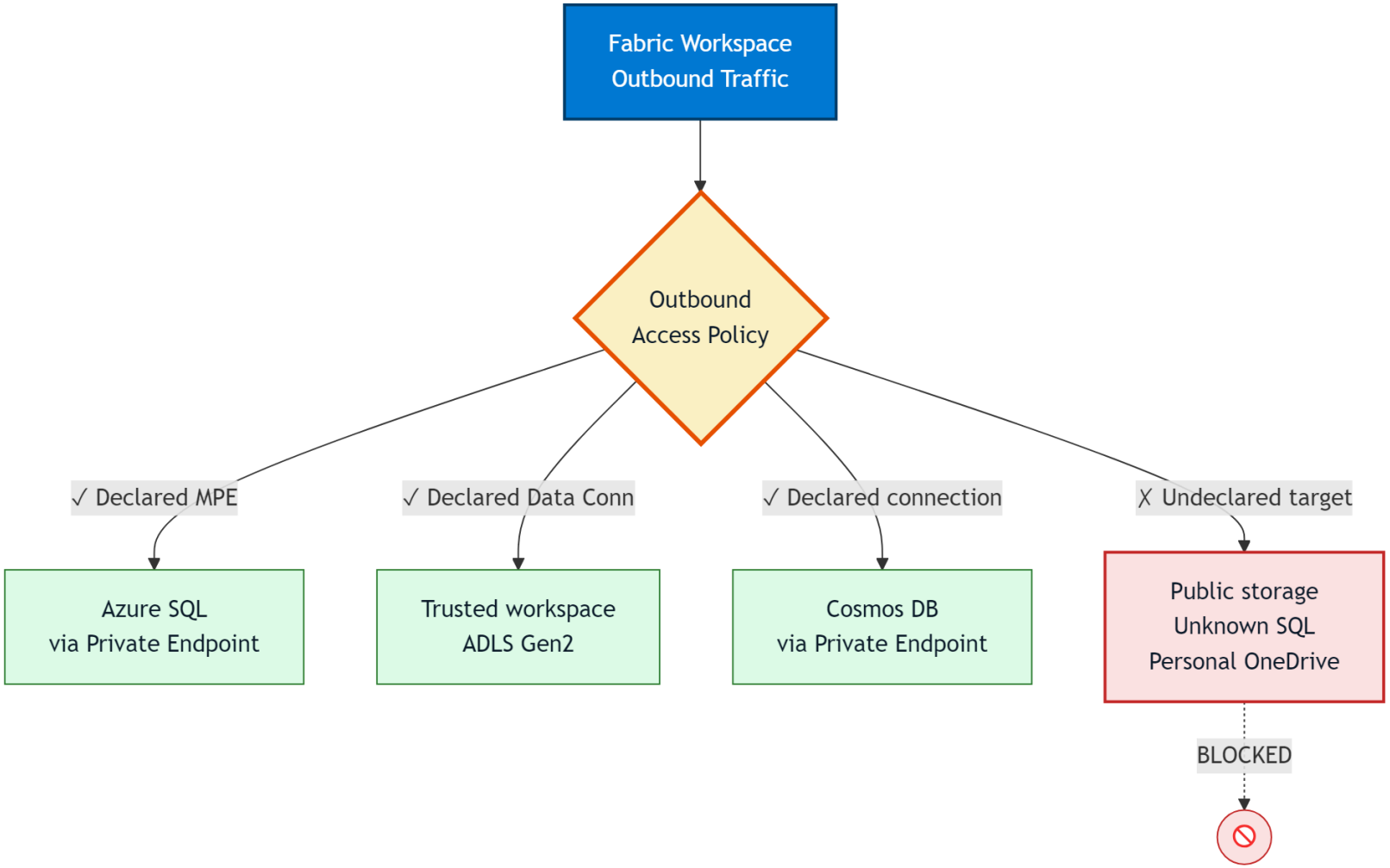
Heuristic. Azure PaaS → MPE. On-prem → On-Prem GW. Your VNet → VNet GW. ADLS only → TWA.

03

Exfiltration Protection.

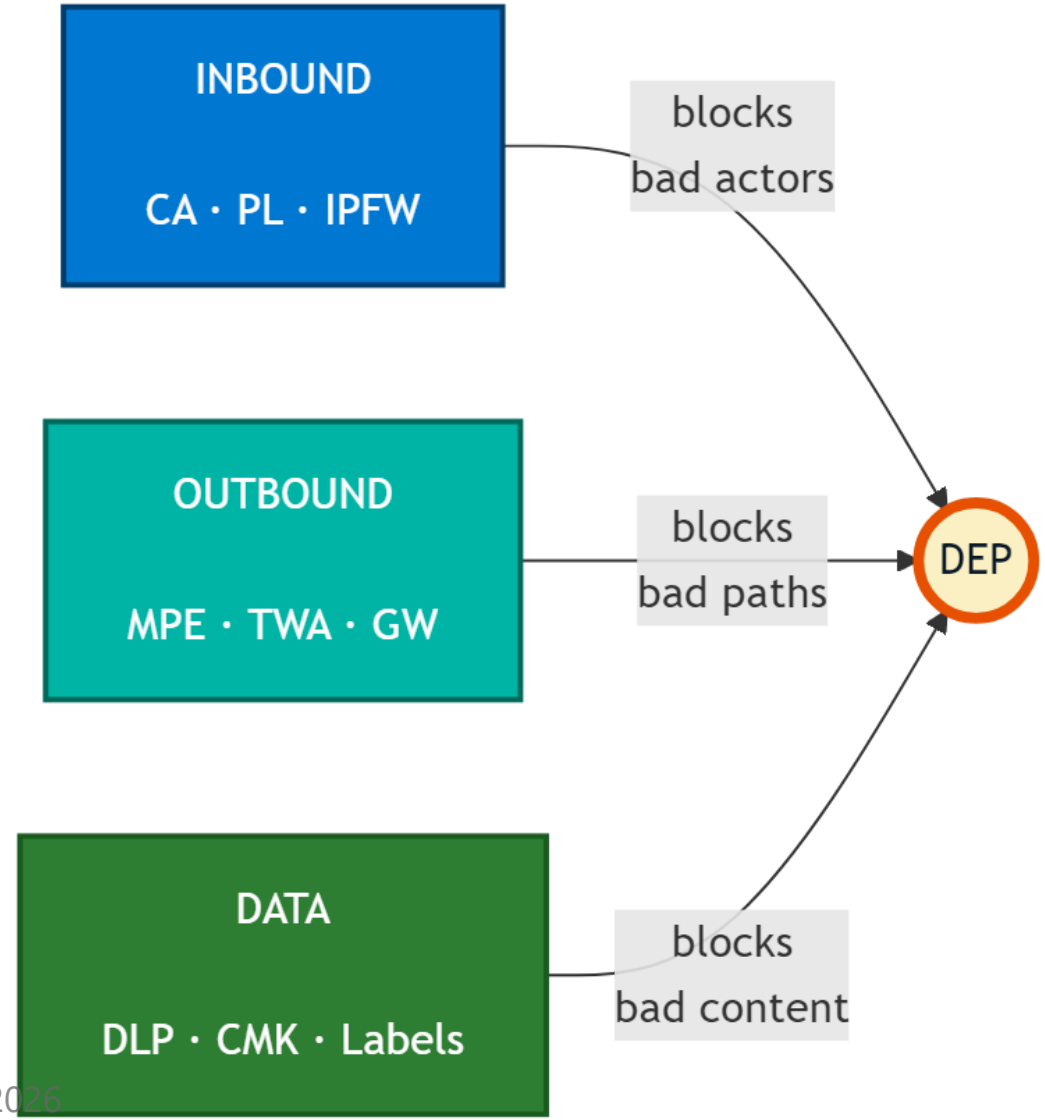
CONTROLLING WHERE DATA GOES — NOT JUST HOW IT GETS OUT.

Outbound Access Policies



Traffic must match a **declared destination** — MPE or Data Connection. Anything else: blocked by default.

The Full DEP Picture



Data Exfiltration Protection isn't a single feature — it's the **convergence of three layers**.

- **Inbound** blocks bad actors at the door
- **Outbound** blocks bad paths to the outside
- **Data** blocks bad content even on authorized paths

Remove any one layer and DEP leaks. Network alone won't stop an authenticated user emailing a spreadsheet. Labels alone won't stop a rogue notebook posting to an unknown endpoint.

Content Controls · Layer 3

EXPORTS

Power BI Export Restrictions

Disable Excel / CSV / PPTX exports on sensitive workspaces.

ENDPOINT

Endpoint DLP

Purview + Intune to block copy to USB or personal cloud.

CLASSIFY

Sensitivity Labels

Auto-apply via Purview — labels travel with the data.

ENCRYPT

Customer Managed Keys

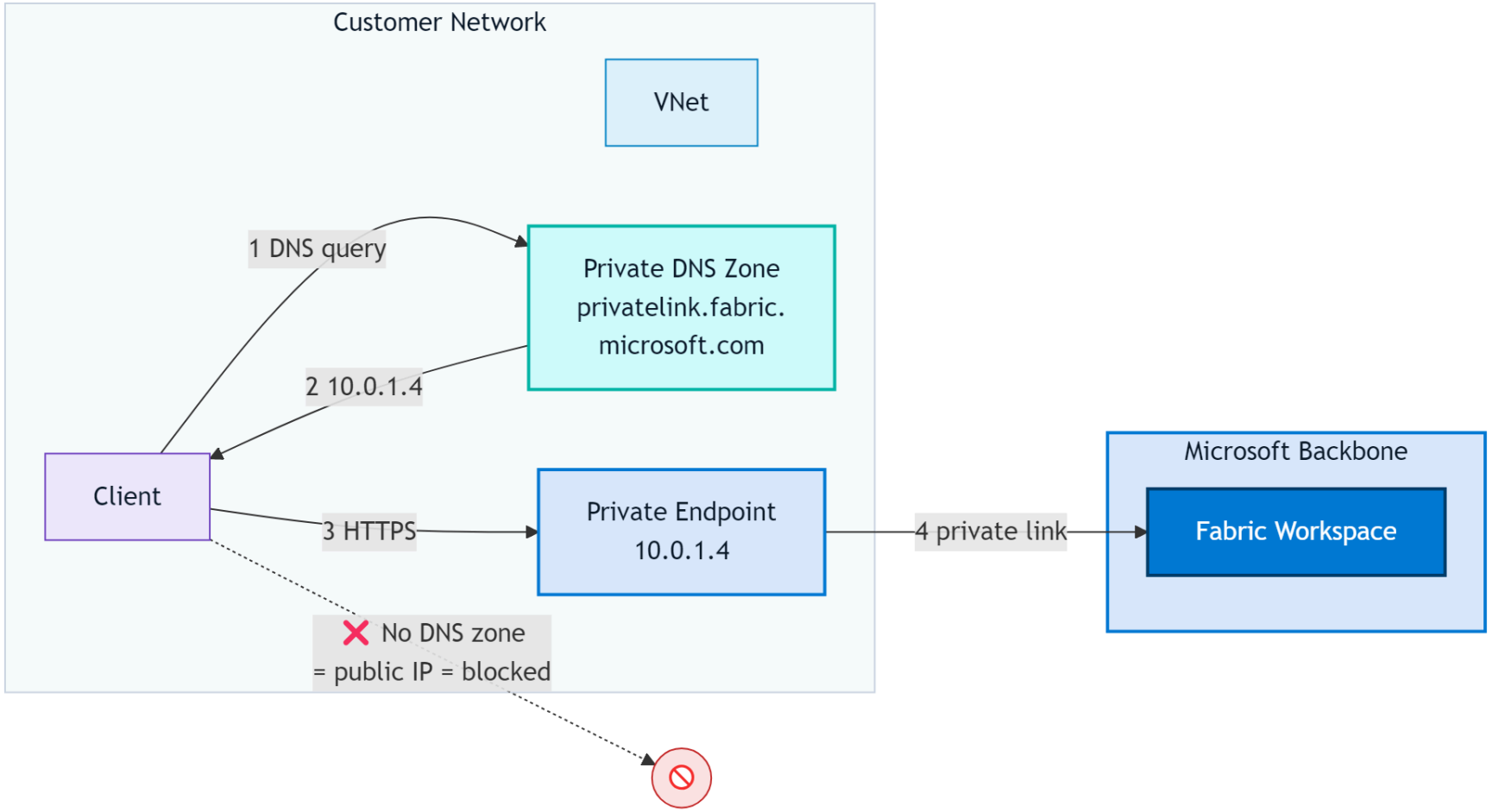
Additional encryption boundary with full key control.

04

Operations.

DNS, MONITORING, TESTING — THE LAYERS THAT MAKE IT ACTUALLY WORK.

DNS · The Silent Killer



Without the Private DNS Zone, clients resolve the public IP — Private Link then blocks. The result: **"works from one machine but not another."**

Monitoring Stack

DIAGNOSTICS

Log Analytics

Access patterns, query perf, errors.

NETWORK

Network Watcher

VNet flows, connectivity tests, PE reachability.

SECURITY

Azure Sentinel

SIEM correlation. Anomalous access, failed CA.

COMPLIANCE

Microsoft Purview

Classification, DLP, sensitivity audit.

FIREWALL

Azure Firewall

Centralized egress logging. Static egress IP.

AUDIT

Quarterly review

Rotate IPs, re-validate CA policies.

Validation Checklist

01 Resolve DNS from a client

`nslookup workspace.fabric.microsoft.com` must return a **private IP (10.x)**.

02 Test blocked public access

From a non-allowed IP the connection must fail explicitly.

03 Spark cold start

First Spark job activates the Managed VNet — measure latency.

04 MPE approval flow

Verify the target service admin receives and can approve the request.

05 Exfiltration attempt

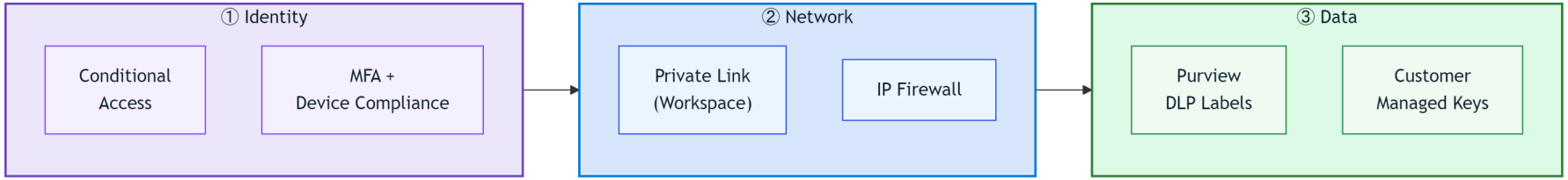
Write to an undeclared destination. The policy must block it.

05

Recommendations.

WHAT WOULD I DEPLOY IN YOUR CONTEXT? SCENARIO-DRIVEN BLUEPRINTS.

Zero Trust · Reference Posture



Identity is the primary perimeter. Network is the second. Data is the last line.

Architectures by Scenario

STANDARD ENTERPRISE

Balanced · Default choice

CA + Workspace PL + MPE + Outbound Policies.

REGULATED · GDPR · PCI

Maximum Control

CA + WS PL + MPE + Outbound Policies + **Purview DLP** + **CMK**.

HYBRID DATA ESTATE

On-prem integration

CA + IP Firewall + **VNet Gateway** + **On-Prem Gateway**.

MULTI-TEAM ISOLATION

Segregated workloads

CA + **Per-workspace PL** + Per-workspace outbound + **PIM**.

Feature Roadmap

Feature	Status	Notes
Conditional Access	GA	Since day one
Private Link (Tenant)	GA	All-or-nothing
Private Link (Workspace)	GA	**Recommended**
IP Firewall	GA	Early 2026
Managed VNet / MPE	GA	Core outbound
Trusted Workspace Access	GA	ADLS Gen2 only
Outbound Access Policies	GA	End 2025
VNet GW · Cert + Proxy	GA	April 2026
Customer Managed Keys	GA	All workloads
Eventstream Private Network	Preview	Early 2026
Power BI network protection	Planned	Late 2026
Fabric Database network protection	Planned	No ETA

TAKEAWAYS

Three pillars. Defense in depth. Zero blind spots.

START WITH IDENTITY. LAYER NETWORK CONTROLS. FINISH WITH DATA CLASSIFICATION. TEST EVERY LAYER. MONITOR THE WHOLE STACK.

Source: github.com/fredgis/Divers/markdown/Fabric_Network_Security.md